PianoPro: Learning a Dexterous Piano Player By RL and Conditioned Diffusion

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Abstract

Robotic piano performance presents a formidable challenge, requiring dexterous fingertip control and precise temporal coordination. Building upon previous methods such as PianoMime, which use imitation learning and PPO-based agents, we propose PianoPro: a unified framework for learning expressive and physically realistic robotic piano performances. PianoPro introduces three key innovations: (1) Dynamic Reward Modulation, which gradually shifts training focus from trajectory mimicry to task accuracy; (2) Extended Audio-Based Music Reward, which evaluates generated performances using both waveform similarity and MFCC-based perceptual metrics; and (3) Fingering-Guided Conditional Diffusion, which augments trajectory generation with human-inspired fingering heuristics to produce biomechanically plausible motions. Experiments on dexterous piano playing tasks demonstrate that PianoPro achieves superior F1 scores and audio fidelity compared to previous methods, yielding more natural, accurate, and expressive robot performances.

1 Introduction

Robotic piano performance represents a significant challenge in dexterous manipulation, requiring precise fingertip control and millisecond-level temporal accuracy for expressive musical execution. Our work builds upon PianoMime [1], which employs a three-stage methodology: (1) extracting expert fingertip trajectories from human performances; (2) training PPO-based specialized agents to replicate these trajectories using mimicry and residual learning; and (3) learning a generalist policy via imitation learning. This generalist policy leverages a two-level control hierarchy: high-level conditional diffusion models generate temporally aligned fingertip trajectories from symbolic MIDI, while low-level inverse dynamics models translate them into joint actions.

While this pipeline achieves promising keypress accuracy, it exhibits several limitations. First, the reward function is rigid—fixed-weight mimicry objectives hinder the adaptive balance between imitation fidelity and task performance. Second, evaluation metrics such as precision and recall fail to reflect perceptual and auditory aspects like realism and musicality. Third, conditioning solely on MIDI omits valuable fingering priors, leading to biomechanically implausible motions and reduced execution feasibility.

To address these challenges, we introduce **PianoPro**, a unified framework that incorporates three key innovations: (1) **Dynamic Reward Modulation** that transitions the learning focus from trajectory mimicry to task accuracy over time; (2) **Extended Audio-Based Music Reward** that evaluates generated audio using waveform similarity and MFCC-based perceptual metrics; and (3) **Fingering-Guided Conditional Diffusion**, which augments the trajectory generation process with human-inspired fingering heuristics. Together, these improvements yield more accurate, expressive, and physically realistic performances, achieving higher F1 scores and audio fidelity than prior methods.

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2 Related Work

Robotic piano performance represents a complex intersection of dexterous manipulation, real-time planning, and musical expression. Recent research has evolved along two primary trajectories: autonomous piano playing systems and human-robot collaborative frameworks.

In the domain of autonomous piano performance, significant advances have been achieved through reinforcement learning paradigms. Zhao et al.'s RP1M [2] establishes a foundational approach by curating a large-scale dataset of approximately one million expert trajectories across two thousand musical pieces. A key innovation in this work is the formulation of fingering optimization as an Optimal Transport problem, solved efficiently via the Hungarian algorithm to minimize cumulative finger travel distance without relying on human annotations. This geometric formulation is integrated into a multi-term reward function combining key pressing accuracy, sustain pedal control, collision avoidance, and energy efficiency constraints. The framework employs domain randomization techniques to bridge the sim-to-real gap and demonstrates generalization capabilities through hybrid execution modes that combine simulated joint states with real-world key sensing.

Building upon prior advances in autonomous piano playing, some efforts have advanced both autonomous dexterous play and realtime human–robot duet systems. RoboPianist [3] taught simulated anthropomorphic hands to execute 150 complex pieces via deep RL, introducing human fingering priors to tackle exploration in high-dimensional control tasks . More recently, PANDORA [4] has innovated with diffusion policy learning: it iteratively denoises noisy action sequences into smooth trajectories, optimizing a composite reward that blends key accuracy, audio fidelity, and semantic feedback from an LLM oracle — attaining new benchmarks in expressiveness and precision within the RoboPianist environment.

On the collaborative front, [5] introduced an RNN-based improvisation model for chord prediction, paired with an MPC adaptive controller for temporal synchrony. Tested in both simulation and hardware "duet" with a custom anthropomorphic arm, the system achieved a 93% chord-prediction accuracy and reliable homophonic accompaniment

3 Method

3.1 Dynamic Reward Modulation

Previous approaches typically incorporate a fixed-weight mimicry reward to encourage human-like behavior in policy learning. This reward serves two primary purposes: (i) to accelerate early-stage convergence by guiding the agent toward expert trajectories, and (ii) to act as a soft constraint, discouraging unnatural or physically implausible keypress postures that may nonetheless yield high task rewards.

In our work, we propose a **dynamic reward modulation** mechanism that adaptively adjusts the influence of the mimic reward over the course of training. During the early phase, a higher weight is assigned to the mimicry component, enabling the policy to rapidly converge toward human-like keypress patterns. As training progresses and the policy approaches a reasonable estimate of human motion, the weight is gradually reduced, allowing the agent to shift its focus toward optimizing task-specific performance, particularly keypress accuracy.

Formally, the reward weights are defined as:

$$\begin{cases} \omega_{\text{Key Press}}(t) = 1 - mq^t \\ \omega_{\text{Mimic}}(t) = mq^t \end{cases} \tag{1}$$

where t denotes the training timestep, $m \in [0, 1]$ denotes the initial weight for mimic reward, and $q \in [0, 1]$ is a decay factor controlling the transition rate between imitation and task optimization. $\omega_{\text{Key Press}}(t)$ increases over time to emphasize task reward, while $\omega_{\text{Mimic}}(t)$ decreases to reduce reliance on mimicry reward.

This dynamic reward scheme offers three key advantages:

1. **Improved final performance:** By focusing more on task accuracy in the later stages, the final policy achieves a higher F1 score with more precise keypress execution.

Table 1: The detailed reward function to train the song-specific policy. The Key Press reward and the Mimic reward are the same as Pianomime [1], where k_s and k_g represent the current and the goal states of the key respectively, and g is a function that transforms the distances to rewards in the [0, 1] range. p_{df} and p_{rf} represent the fingertip positions of human demonstrator and robot respectively.

Reward	Formula	Weight	Explanation
Key Press	$0.5 \cdot g(\|\|k_s - k_g\|\|_2) + 0.5 \cdot (1 - \mathbf{1}_{\text{false positive}})$	$1 - mq^t$	Press the right keys and only the right keys
Mimic	$g(\|\|p_{df} - p_{rf}\|\|_2)$	mq^t	Mimic the demonstrator's fingertip trajectory

2. **Faster convergence:** The early emphasis on mimicry helps the policy quickly approach a meaningful solution space close to human demonstrations.
3. **Balanced learning:** The decay mechanism enables the policy to maintain natural, human-like hand postures while gradually improving task-specific performance.

3.2 Extended Audio-Based Music Reward

3.2.1 Traditional Metrics

Traditional evaluation of piano performance commonly relies on discrete metrics such as precision, recall, and F1-score. For a sequence of key presses, these metrics are defined as follows:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \tag{2}$$

where TP, FP, and FN denote the number of true positives, false positives, and false negatives in key press detection, respectively.

3.2.2 Audio-Based Performance Evaluation

Although discrete metrics offer a basic assessment of accuracy, they fail to capture the continuous and temporal nuances inherent in musical expression. To address this limitation, we introduce a comprehensive audio-based evaluation framework that incorporates multiple levels of analysis.

MIDI-to-Audio Synthesis. The first step converts MIDI sequences into audio waveforms using a physics-based piano synthesizer. For both the target and robotic performances, MIDI events are sampled at a fixed timestep $\Delta t = \frac{1}{30}$ seconds, with a constant velocity ($v = 64$) and sampling rate ($f_s = 44100$ Hz).

Let $\mathbf{M}_t$ and $\mathbf{M}_r$ represent the MIDI sequences for the target and robot performances, respectively: $\mathbf{M}_* = \{(k_i, t_i, d_i, v_i)\}_{i=1}^N$ where k_i is the MIDI pitch number (ranging from 21 to 108 for a standard piano), t_i is the onset time, d_i is the duration, and v_i is the velocity (fixed at 64). These sequences are converted into audio waveforms $\mathbf{a}_t, \mathbf{a}_r \in \mathbb{R}^T$ via a synthesis function $\mathcal{S}$: $\mathbf{a}_* = \mathcal{S}(\mathbf{M}_*, f_s)$

Multi-Level Similarity Analysis. The evaluation framework combines two complementary perspectives:

1. **Waveform Similarity:** We compute the cosine similarity between the raw audio waveforms:
$$S_{\text{wave}} = \frac{\mathbf{a}_r \cdot \mathbf{a}_t}{\|\mathbf{a}_r\| \|\mathbf{a}_t\|} \tag{3}$$
2. **Spectral Features (MFCC):** Mel-frequency cepstral coefficients (MFCCs) are extracted through the following procedure:
   - *Short-Time Fourier Transform (STFT)*:
$$X[k, m] = \sum_{n=0}^{N-1} w[n] x[n + mH] e^{-j2\pi kn/N} \tag{4}$$

- *Mel Filterbank Energy*:

$$E[m,l] = \sum_{k=0}^{N/2} |X[k,m]|^2 H_l[k] \tag{5}$$

- *Log Compression and Discrete Cosine Transform (DCT)*:

$$c[m,i] = \sum_{l=0}^{L-1} \log(E[m,l]) \cos\left(\frac{\pi i(l+0.5)}{L}\right) \tag{6}$$

This results in MFCC feature matrices $\mathbf{M}_t, \mathbf{M}_r \in \mathbb{R}^{T\times D}$. And the similarity of these spectral features is computed as:

$$S_{\text{mfcc}} = \frac{1}{T}\sum_{t=1}^{T} \frac{\mathbf{M}_r[t]\cdot\mathbf{M}_t[t]}{\|\mathbf{M}_r[t]\|\|\mathbf{M}_t[t]\|} \tag{7}$$

The final audio similarity score is a weighted combination of the waveform and MFCC similarities:

$$S_{\text{audio}} = \alpha_1 S_{\text{wave}} + \alpha_2 S_{\text{mfcc}}, \quad \text{with } \alpha_1 + \alpha_2 = 1 \tag{8}$$

Implementation Details. To enable real-time feedback during training, we compute these metrics over sliding windows. The evaluation window centered at time t is defined as:

$$w(t) = \left[t - \frac{W}{2}, t + \frac{W}{2}\right] \tag{9}$$

where W is the window width (typically set to 0.5 seconds). The full set of metrics at each training step t is given by:

$$\mathcal{M}_t = \{F1_t, Recall, Precision, S_{\text{audio},t}\} \tag{10}$$

3.3 Fingering-Guided Conditional Diffusion

In prior work such as PianoMime [1], the diffusion-based policy is conditioned only on desired musical notes (e.g., MIDI sequences), generating high-level fingertip trajectories without considering detailed biomechanical constraints. However, this conditioning scheme overlooks a critical cue in human piano playing: the real-time configuration and activity of individual fingers.

To address this limitation, we propose a fingering-guided conditional diffusion model. Specifically, we introduce a 10-dimensional fingering feature vector $\mathbf{f} \in \mathbb{R}^{10}$, where each entry corresponds to the normalized actuation level of a finger (5 per hand). A value of 1 indicates that the finger is actively pressing a key, while -1 denotes inactivity. This fingering vector captures instantaneous hand posture and is temporally aligned with the target trajectory segment.

We incorporate this fingering cue as an additional condition into the diffusion denoising network. Formally, the model learns to predict the clean trajectory $\mathbf{x}_0$ from a noisy sample $\mathbf{x}_t$ at timestep t, given both the musical goal and the current fingering vector:

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{\mathbf{x}_0,\epsilon,t}\left[\|\epsilon - \epsilon_\theta\left(\mathbf{x}_t, t \mid \mathbf{M}, \mathbf{f}\right)\|^2\right] \tag{11}$$

The denoising network ϵ_θ is implemented as a Transformer-based architecture, where the fingering vector $\mathbf{f}$ is projected and concatenated with the timestep embedding and MIDI encoding at each layer. This design allows the model to leverage finger-specific activation patterns to generate more accurate and physically plausible motion plans.

Benefits. Incorporating fingering information yields the following advantages:

- **Physical Plausibility:** Fingering-aware conditioning encourages more natural and physically feasible trajectories that better match the capabilities of human or robotic hands.
- **Search Space Reduction:** Fingering priors effectively constrain the high-level diffusion model to a smaller subspace of plausible behaviors, accelerating convergence and improving sample efficiency.
- **Improved Control Quality:** The resulting trajectories exhibit better temporal continuity and ergonomic consistency, which in turn facilitate more accurate downstream joint-level control via inverse dynamics.

Overall, our fingering-guided conditional diffusion framework infers fingertip motions that are not only musically correct but also aligned with human fingering conventions, leading to more expressive and executable robot performances.

4 Experiments

We conduct comprehensive experiments to evaluate the effectiveness of the proposed PianoPro framework. For fair comparison and reproducibility, we adhere closely to the experimental setup of PianoMime [1], including dataset, preprocessing, and core model architecture. Below, we outline the experimental details, emphasizing key differences introduced in our implementation.

4.1 Setup and Dataset

We use the same dataset and preprocessing pipeline as PianoMime [1], ensuring a consistent foundation for comparison. The core architecture of our models also follows the original design, including both the high-level trajectory generator and the low-level inverse dynamics controller.

4.2 Implementation Details

All experiments are conducted using a single NVIDIA GeForce RTX 3090 GPU with PyTorch 2.4.0 and CUDA 12.4. Our training pipeline is structured into two stages: specialized agent training and generalized policy learning.

Specialized Agent Training. To improve training efficiency and reward dynamics, we modify the original setup in the following ways:

- **Virtual environments:** Increased from 32 to 64 to accelerate sampling.
- **Mimic reward schedule:** Starts at 0.67 with an exponential decay factor of $q = 0.999$ to gradually shift emphasis from imitation to task accuracy.
- **Audio reward:** Introduced with a constant weight of 0.3 to promote perceptual alignment with human performances.

Generalized Policy Learning. In this stage, we incorporate fingering priors into the conditional diffusion model to guide trajectory generation. All other training and evaluation settings remain consistent with those used in PianoMime [1].

5 Results

Dynamic Reward Modulation The proposed dynamic reward modulation mechanism demonstrates significant performance improvements across all evaluation metrics. As shown in Figure 1, our approach achieves a final F1 score of 92.38%, representing a 0.3% improvement over the baseline. This enhancement stems directly from the time-varying reward strategy: during early training (first 20% of iterations), the strong mimicry focus ($\omega_{\text{Mimic}} \approx 0.67$) rapidly guides policies toward human-like hand postures. As training progresses, the exponential decay ($q = 0.999$) systematically shifts emphasis toward task accuracy, enabling policies to refine keypress precision without sacrificing motion naturalness.

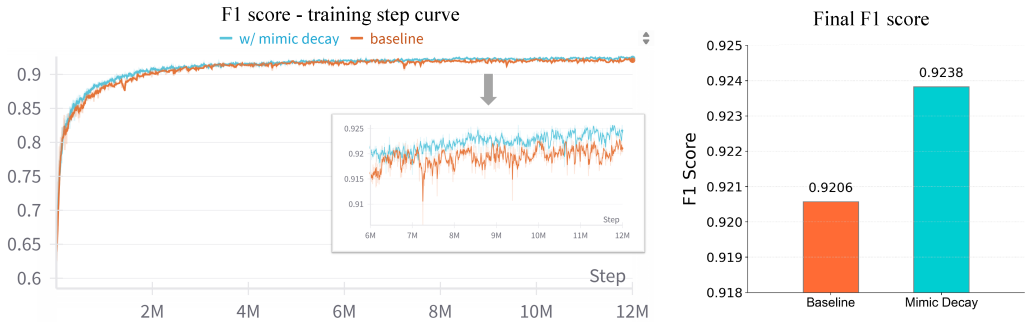

Figure 1: Comparison between F1 scores of baseline and our method

Figure 2: Qualitative comparison of finger postures: (Left) Our dynamic reward maintains anatomically plausible configurations throughout chord transitions; (Right) .

Visual comparisons reveal profound differences in finger kinematics: Figure 2 demonstrates how the policy without mimic reward frequently produce anatomically implausible configurations (e.g., extreme finger hyperextension or cross-over collisions), whereas our dynamic approach maintains natural human-like postures throughout performances. This visual evidence confirms that the time-varying reward strategy preserves ergonomic integrity while optimizing accuracy.

Extended Audio-Based Music Reward After applying implicit mimic weight decay, the audio score increased to 93.42, compared to 93.01 for the baseline. Furthermore, we investigated the effect of incorporating an audio-based reward during training. The results are presented in Figure 3.

Due to the reduced training speed after introducing the audio reward, we adopted a two-stage strategy in our experiments: first training the model with the standard baseline loss, followed by fine-tuning using the audio reward. The F1 score after fine-tuning with audio reward remained essentially consistent with the baseline (92.52% versus 92.34%), indicating that introducing the audio reward does not compromise overall prediction accuracy. In contrast, the smoothed audio score curve for the fine-tuned model is consistently higher than that of the baseline throughout the training process, with the best audio score reaching 93.61%—representing a 0.4% absolute improvement over the baseline.

This demonstrates that incorporating audio-based reward provides a clear advantage in terms of audio-related performance and model generalization, without sacrificing the core predictive ability as measured by F1 score.

Fingering-Guided Conditional Diffusion As shown in Figure 4, the fingering-guided conditional diffusion model exhibits both faster convergence and lower final loss compared to the baseline. The additional fingering input helps constrain the solution space and provides informative inductive bias, leading to more sample-efficient learning and improved generation quality.

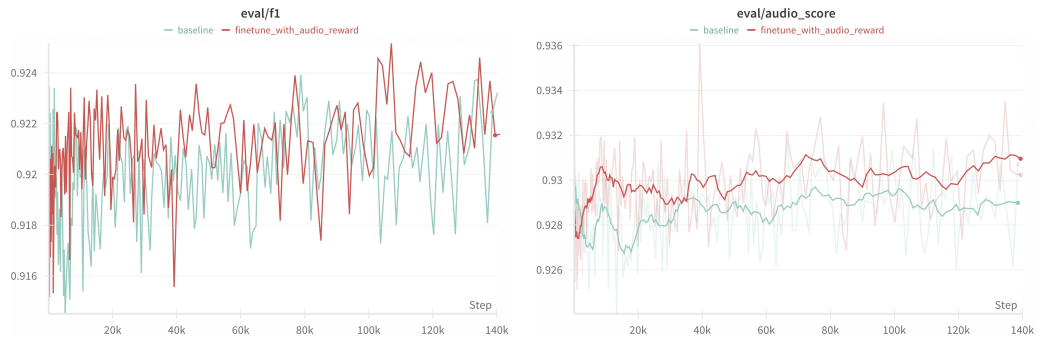

Figure 3: Comparison of F1 score and audio score between the baseline and fine-tuning with audio reward. The darker-colored curve on the right represents the smoothed audio score, which better illustrates the overall trend.

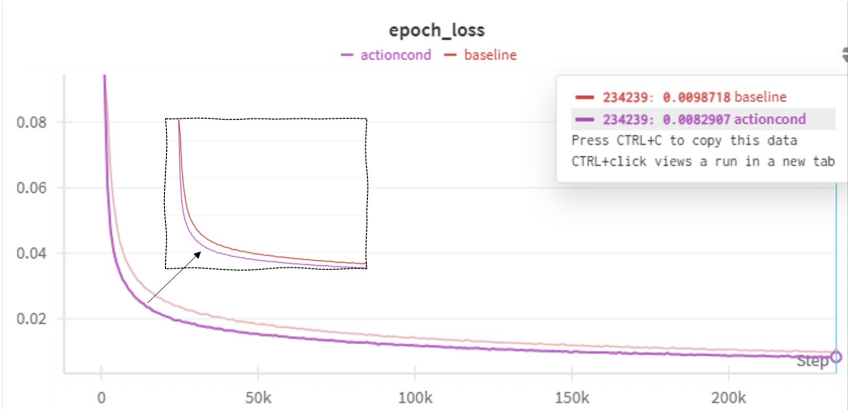

Figure 4: Training loss curves of the baseline model (red) and our fingering-conditioned model (purple).

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